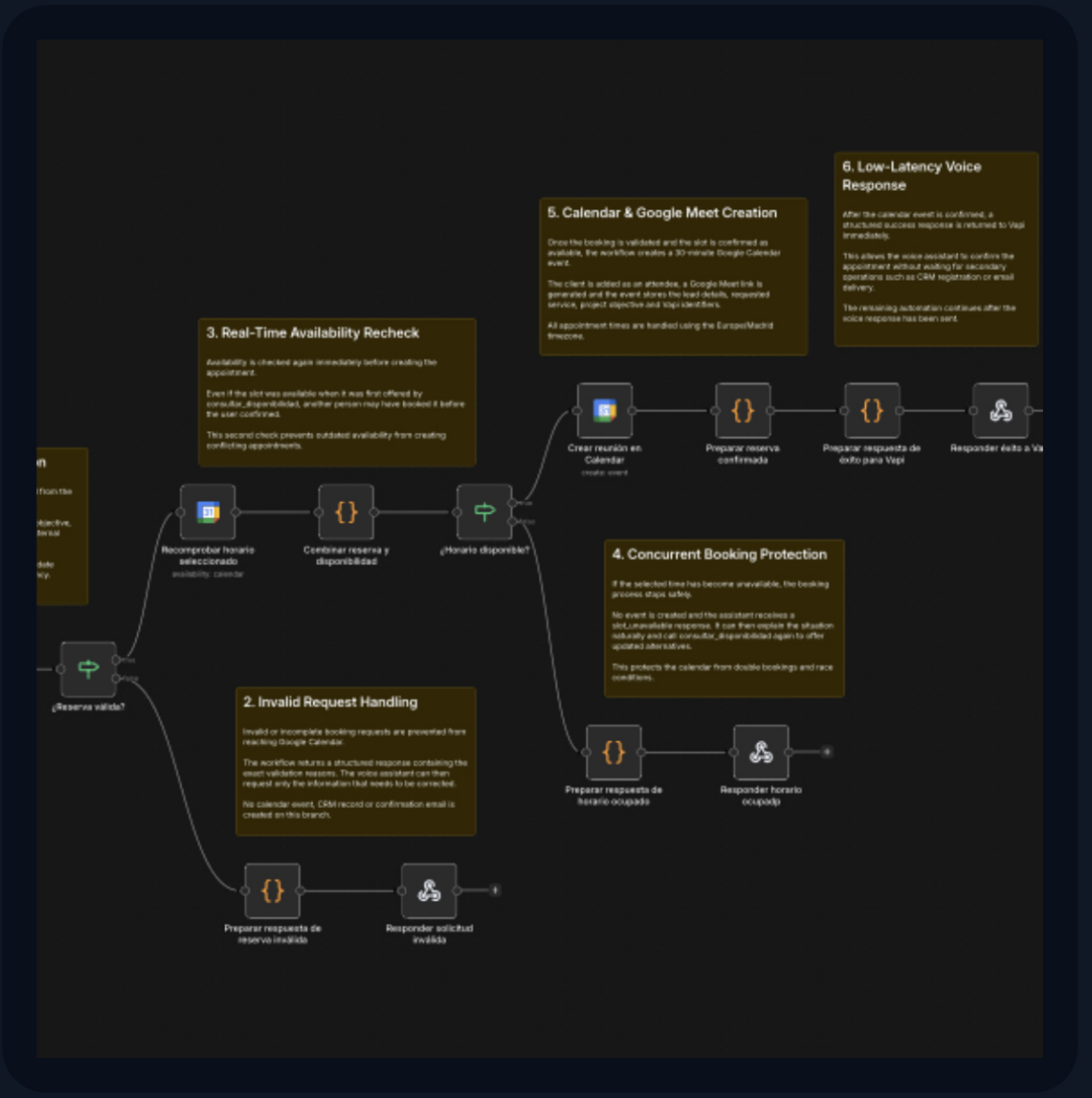
PORTFOLIO CASE STUDY

# AI Voice Appointment Booking & CRM Automation

A Spanish-speaking voice agent that qualifies leads, checks live availability, reduces stale-slot booking risk through a final recheck, creates Google Meet appointments and synchronizes CRM and email follow-up.

Vapi | ElevenLabs | n8n | Google Calendar | Google Sheets | Gmail




Independent demonstration project - built, tested and documented end to end

**Xabier Garcia Sande**

AI Automation & n8n Workflow Development

# Business problem and solution

A production-style demonstration focused on reliable operations, not a scripted chatbot demo.

### Business challenge

Manual qualification and scheduling create delays, context switching and inconsistent follow-up. A slot can also be taken between the first availability check and the final confirmation.

### Automation objective

Let a Spanish voice assistant collect lead context, offer real calendar options, obtain explicit confirmation and complete the operational follow-up automatically.

### Design principle

Separate availability discovery from booking. The final reservation rechecks the chosen slot before creating any event, CRM record or email.

**1 Qualify**

Collect service, objective, identity and business context.

**2 Discover**

Generate candidates and check live Calendar availability.

**3 Confirm**

Present three options and require explicit user approval.

**4 Commit**

Recheck the slot, create Calendar event and Google Meet.

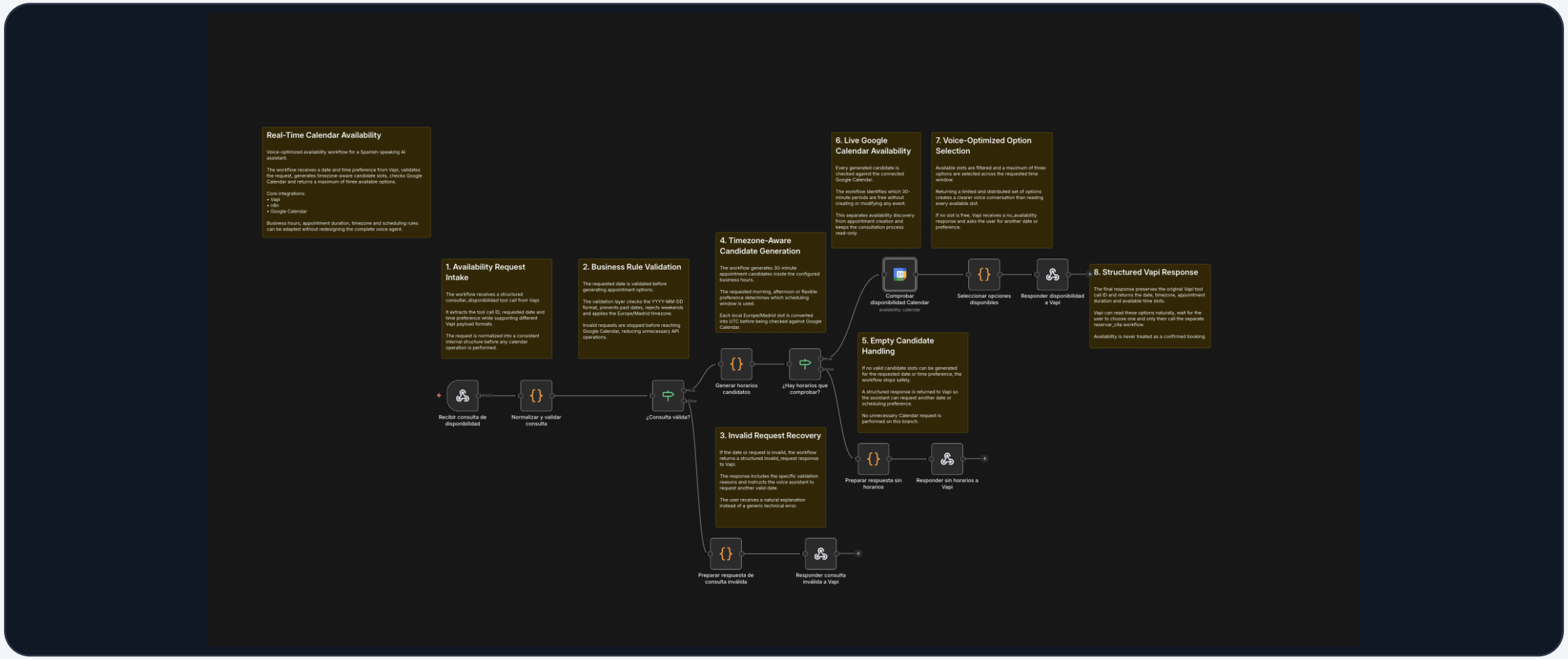
**5 Operate**

Upsert CRM record, send confirmation and track delivery.

Locale: Spanish (Spain) | Business timezone: Europe/Madrid | Meeting duration: 30 minutes

# Read-only calendar availability workflow

The consultar_disponibilidad tool validates the request, generates candidate slots and returns voice-ready options.

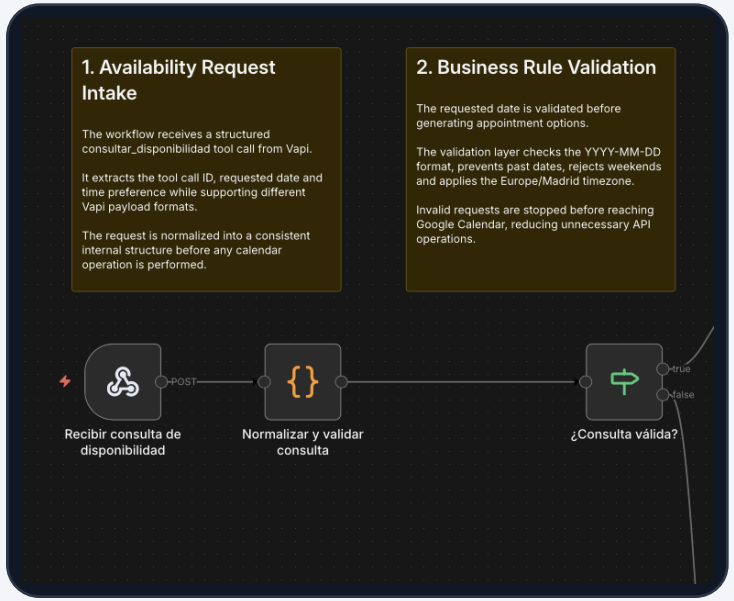


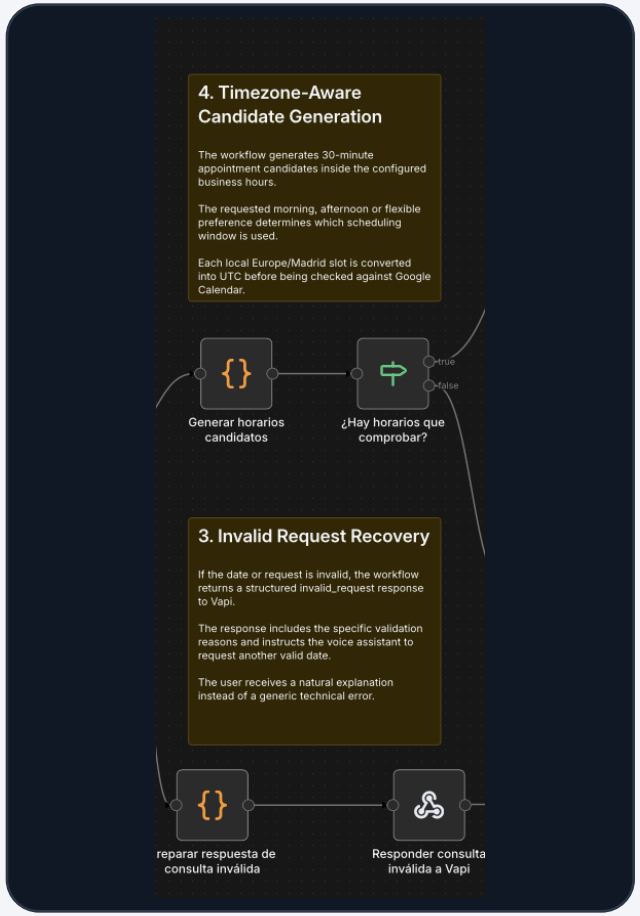

**NO EVENT IS CREATED**

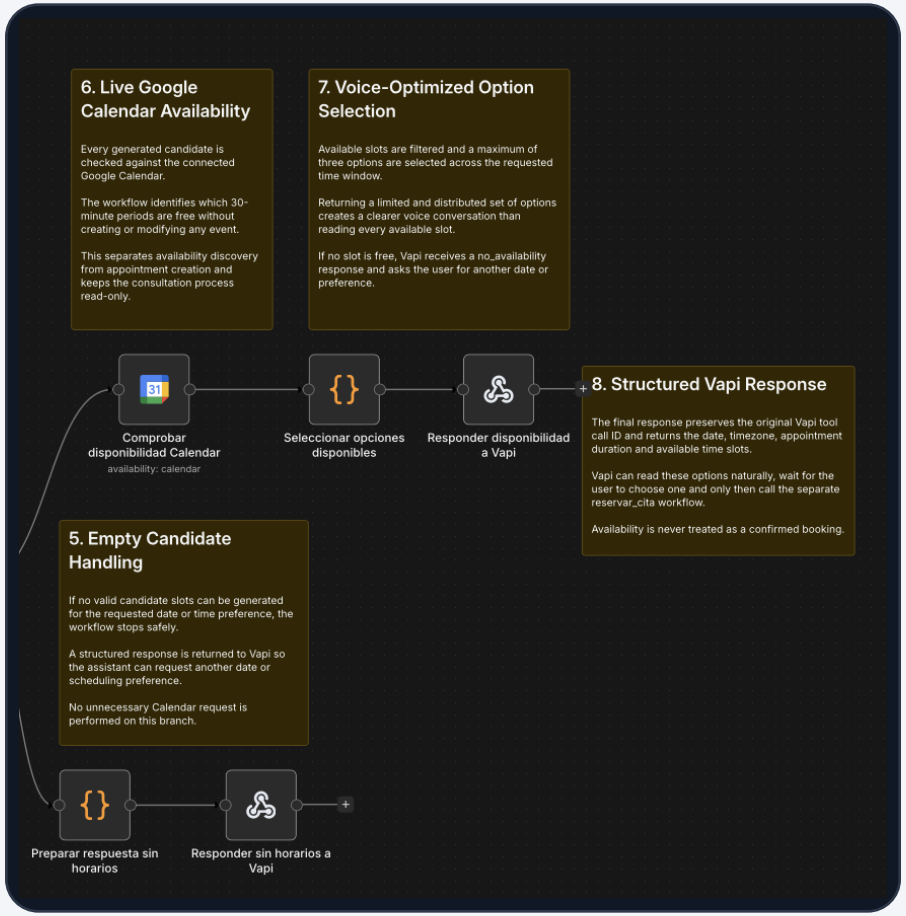
Availability remains a read-only operation until the user explicitly confirms a slot.

# Validation, candidate generation and selection

Each stage limits unnecessary calls and keeps the voice interaction short and understandable.




01 Validate date, weekday and timezone before any API request.

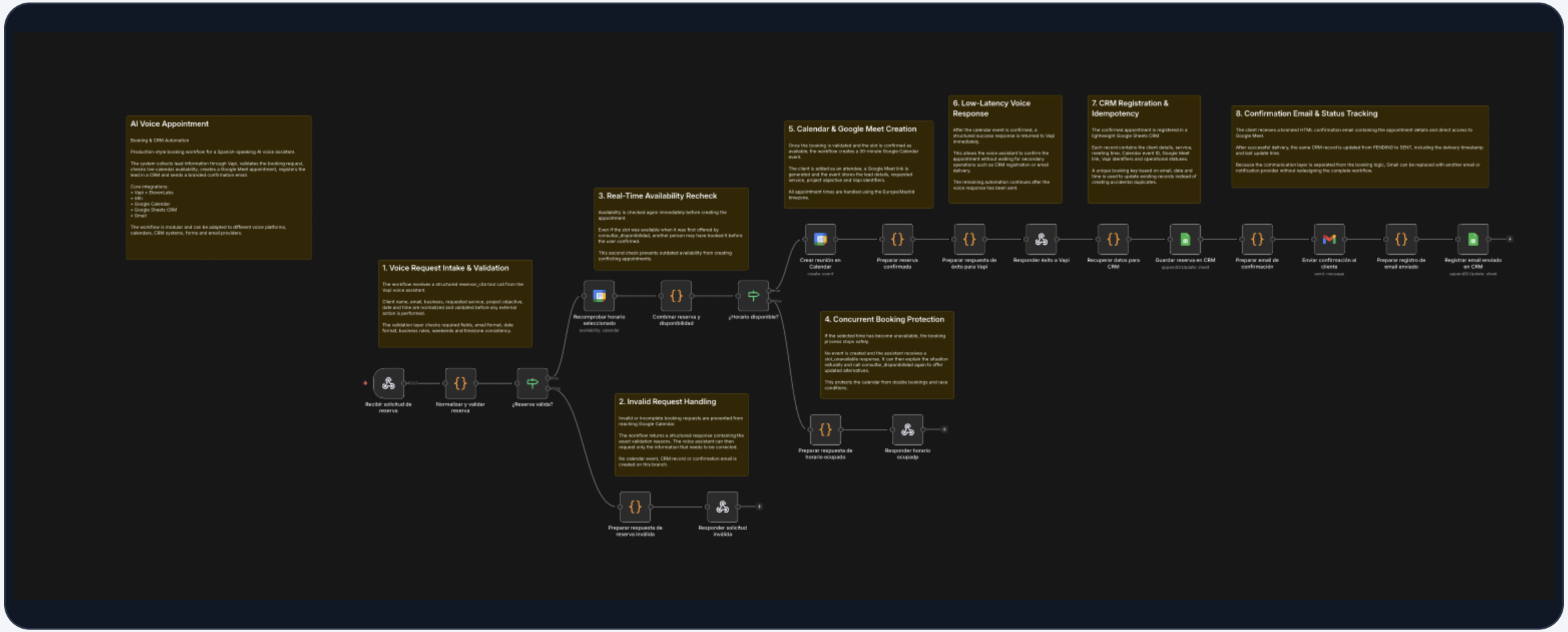
02 Generate 30-minute slots inside configurable business windows.

03 Convert local times to UTC for consistent Calendar checks.

04 Return at most three distributed options for a better voice experience.

# Revalidate, book and complete follow-up

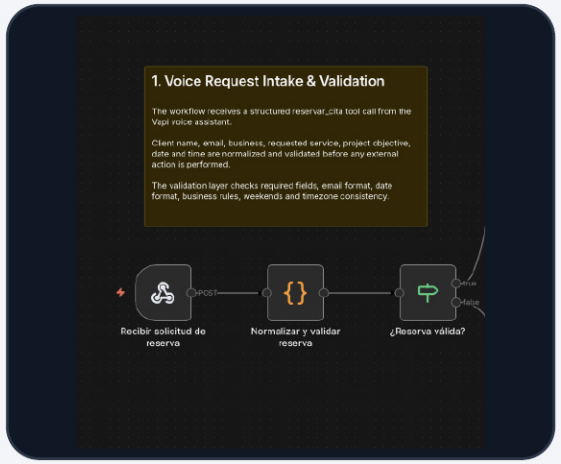
The reservar_cita tool performs the final checks, commits the booking and continues with CRM and email operations.

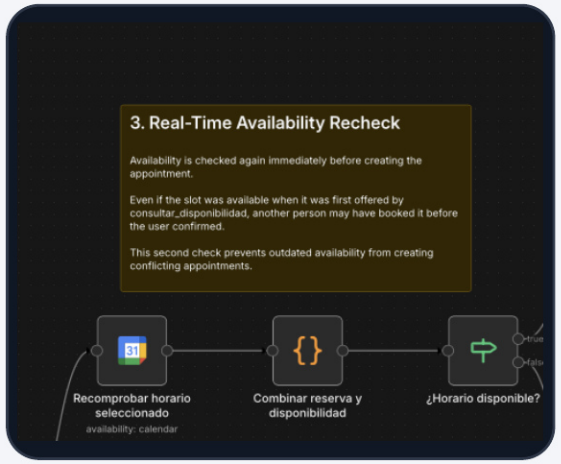


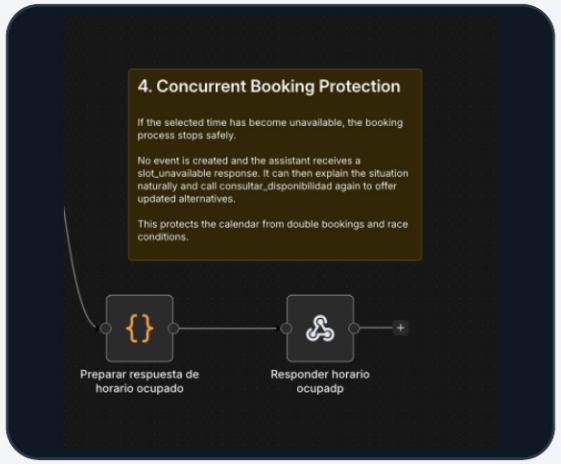

VALIDATE | RECHECK | CREATE MEET | RESPOND TO VAPI | UPSERT CRM | SEND EMAIL | TRACK STATUS

# Guardrails before side effects

Invalid data and newly occupied slots stop before Calendar, CRM or Gmail are allowed to run.







### Input validation

Required fields and booking rules are checked first.

### Availability recheck

The selected slot is checked again at confirmation time.

### Conflict recovery

An occupied slot returns alternatives without booking.

### Calendar commit

Only a valid, free slot creates the event and Meet link.

### Low-latency voice response

Vapi receives the structured booking result immediately after Calendar confirms the event. CRM and email continue afterward.

### Idempotent CRM writes

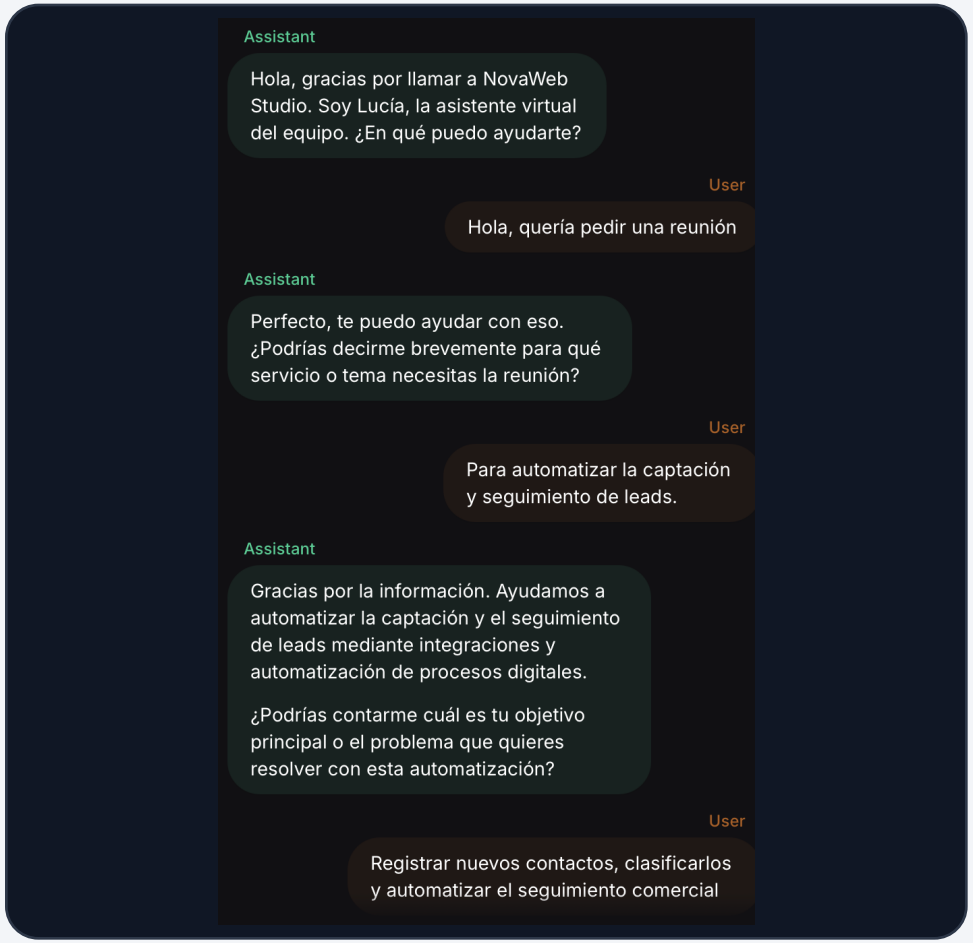
A booking key based on email, date and time updates the matching record instead of appending duplicates.

### Operational status tracking

The CRM records Calendar state, Meet link, Vapi identifiers and confirmation email status with timestamps.

# From business need to a real available slot

The assistant qualifies the request conversationally and uses live Calendar data instead of inventing availability.




**LEAD QUALIFICATION**

Service and project objective are captured before scheduling.



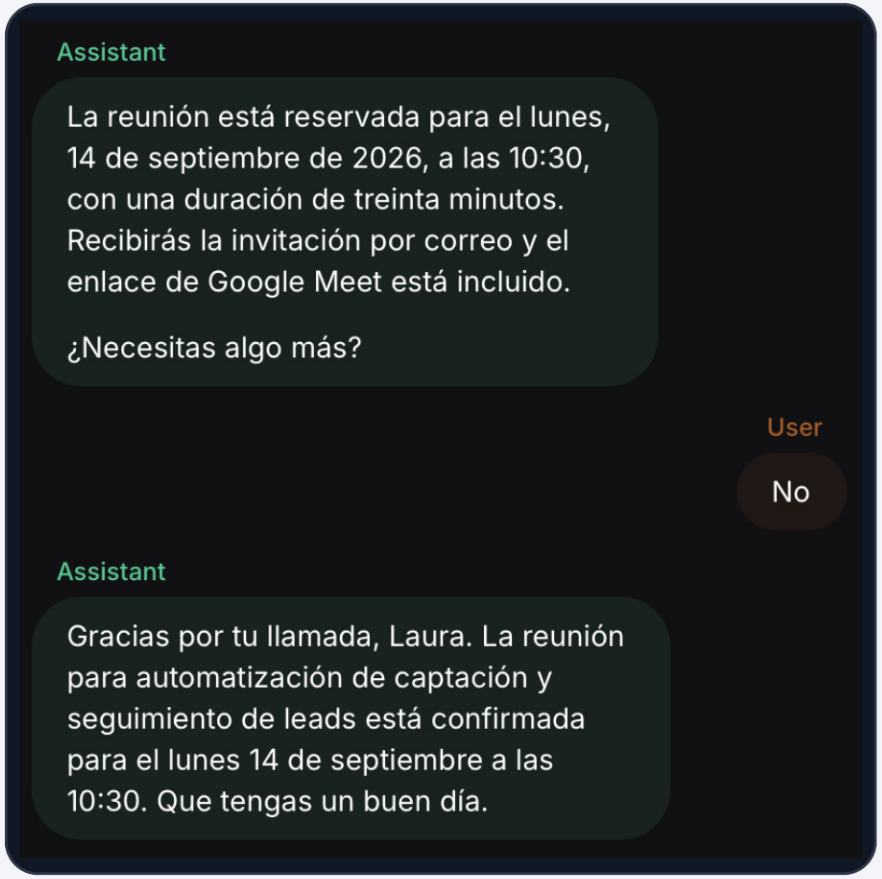

**LIVE AVAILABILITY**

10:00 is rejected and three genuine alternatives are offered.

# No booking without clear user approval

The assistant summarizes every relevant field, waits for a clear yes and only then calls reservar_cita.



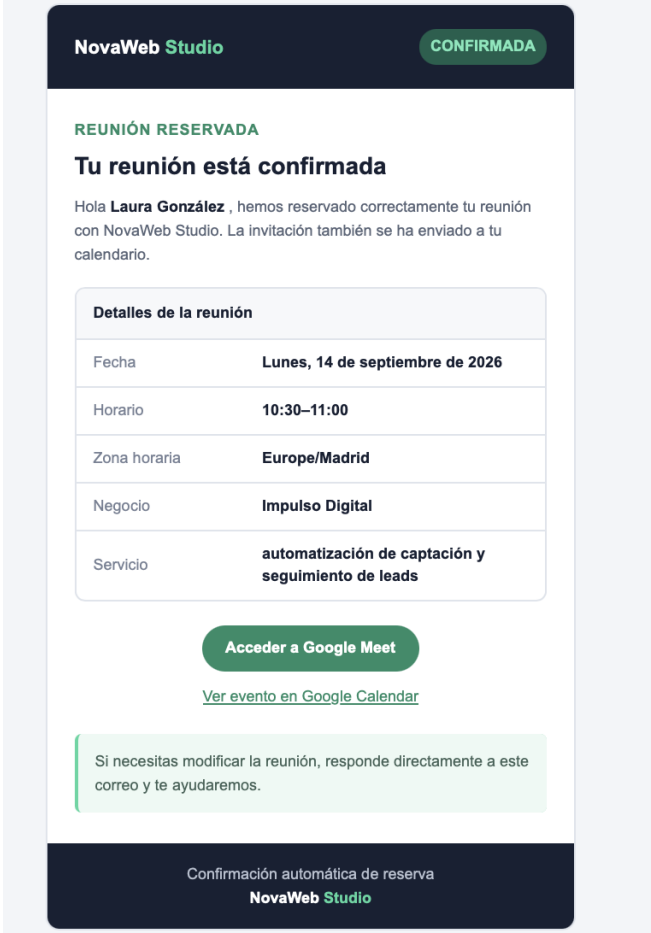




**Result: booked**

The response confirms date, local time, duration, email invitation and Google Meet availability.

# Calendar event and branded confirmation

The automated conversation produces real downstream artifacts, not only a successful tool message.



NovaWeb Studio

CONFIRMADA

REUNIÓN RESERVADA

**Tu reunión está confirmada**

Hola **Laura González** , hemos reservado correctamente tu reunión con NovaWeb Studio. La invitación también se ha enviado a tu calendario.

| Detalles de la reunión | |
|---|---|
| Fecha | **Lunes, 14 de septiembre de 2026** |
| Horario | **10:30–11:00** |
| Zona horaria | **Europe/Madrid** |
| Negocio | **Impulso Digital** |
| Servicio | **automatización de captación y seguimiento de leads** |

Acceder a Google Meet

Ver evento en Google Calendar

Si necesitas modificar la reunión, responde directamente a este correo y te ayudaremos.

Confirmación automática de reserva
NovaWeb Studio

HTML CONFIRMATION EMAIL



Discovery Call — Impulso Digital — Laura González

Lunes, 14 de septiembre · 10:30 – 11:00

Unirse con Google Meet

meet.google.com/***-****-***

1 invitado

1 en espera

x xabi

Cliente: Laura González
Correo: l***@example.com
Negocio: Impulso Digital
Servicio: automatización de captación y seguimiento de leads
Objetivo: Registrar y clasificar nuevos contactos, y automatizar el seguimiento comercial.
Duración: 30 minutos
Vapi Call ID: ••••••••
Vapi Chat ID: ••••••••

CALENDAR + GOOGLE MEET

Personal email, meeting code and internal Vapi identifiers are redacted in the public version.

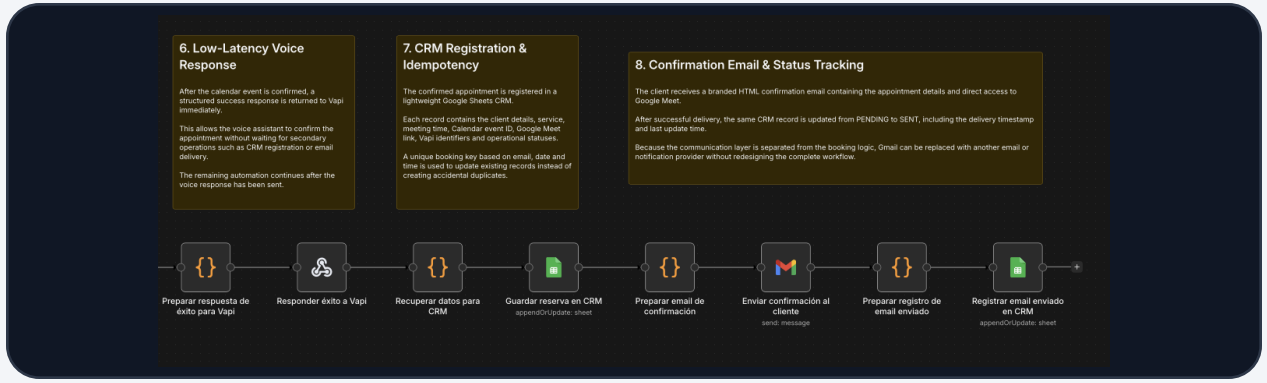
09 / DELIVERY

# CRM traceability, tested branches and adaptability

A sanitized operational record provides end-to-end traceability from voice request to email delivery.

## Sanitized CRM record

| | |
|---|---|
| **Client / business** | Laura Gonzalez / Impulso Digital |
| **Source** | VAPI_VOICE_AGENT |
| **Appointment** | 2026-09-14 \| 10:30-11:00 \| Europe/Madrid |
| **CRM stage** | MEETING_BOOKED |
| **Booking status** | CONFIRMED |
| **Calendar** | Event confirmed \| Meet generated \| Attendee added |
| **Email status** | SENT \| Delivery timestamp recorded |
| **Booking key** | BK-20260914-1030 (public identifier) |




### Three tested decision paths

1. Valid request and free slot: booked successfully.
2. Invalid request: rejected before side effects.
3. Slot occupied after being offered: recheck prevented double booking.

### Portable integration layer

Google Sheets can be replaced with Airtable, HubSpot, Pipedrive or another CRM. Calendar, voice, forms and email providers can also be swapped behind the same normalized data model.

### What this project demonstrates

Business-process analysis, JavaScript data normalization, structured tool responses, timezone handling, API orchestration, human confirmation, conflict protection, CRM idempotency, branded email and auditable statuses.

The same architecture can support web forms, WhatsApp intake, other voice agents or existing client CRMs.

# How the system was built - part I

The project was developed from the conversation design outward, with normalized data between every external service.

### 1 Define the conversation goal

The assistant was designed to qualify a lead and book a discovery call. Required fields were service, project objective, client name, business, email, requested date and preferred time window.

Output: A clear intake checklist and explicit-confirmation rule

### 2 Configure the voice layer

Vapi orchestrates the conversation and ElevenLabs supplies the Spanish voice. The prompt tells the assistant when to ask questions, when to call a tool and when it must wait for the user.

Output: A Spanish-speaking assistant with controlled tool use

### 3 Separate the two tools

consultar_disponibilidad only discovers options. reservar_cita performs the irreversible action. Keeping them separate prevents an availability question from creating an appointment.

Output: Read-only discovery plus controlled booking

### 4 Normalize the Vapi payload

The n8n Code node accepts the different Vapi tool-call payload shapes, parses stringified arguments, preserves the toolCallId and maps aliases into one internal schema.

Output: Stable fields for every downstream node

### 5 Apply business rules early

Dates are checked in YYYY-MM-DD format, past dates and weekends are rejected, and every calculation uses Europe/Madrid. Invalid requests stop before Google Calendar is called.

Output: Structured invalid_request responses without side effects

# How the system was built - part II

Availability, booking and follow-up were implemented as independent stages so each failure path remains predictable.

### 6 Generate and check candidates

The workflow creates 30-minute local slots inside configurable business windows, converts them to UTC and checks each candidate against the connected Google Calendar.

Output: A complete list of genuinely free time slots

### 7 Optimize the voice response

Instead of reading every free slot, the workflow selects up to three options distributed across the requested window. Vapi receives local times, UTC boundaries, date and timezone.

Output: Short, natural choices that are easy to answer

### 8 Recheck before committing

After the user chooses and explicitly confirms a time, reservar_cita checks the exact slot again. If it was taken meanwhile, the workflow returns slot_unavailable and creates nothing.

Output: Final recheck reduces stale-slot booking risk

### 9 Create, respond and continue

A free slot creates the Calendar event, attendee and Google Meet link. Vapi receives the booked result immediately; CRM registration and email delivery continue afterward.

Output: Low voice latency with reliable downstream operations

### 10 Track and verify the outcome

Google Sheets uses a unique booking key for append-or-update behavior. The email status moves from PENDING to SENT, and valid, invalid and newly occupied-slot branches were tested.

Output: Auditable CRM state and three verified decision paths

# Conversation translation - qualification

English translation of the original Spanish Vapi demonstration. Personal contact information remains redacted.

**ASSISTANT**
Hello, thank you for calling NovaWeb Studio. I am Lucia, the team's virtual assistant. How can I help you?

**USER**
Hi, I would like to schedule a meeting.

**ASSISTANT**
Of course. Could you briefly tell me which service or topic you need the meeting for?

**USER**
To automate lead capture and follow-up.

**ASSISTANT**
Thank you. We help automate lead capture and follow-up through integrations and digital process automation. What is your main goal or the problem you want to solve?

**USER**
Register new contacts, classify them and automate the sales follow-up.

**ASSISTANT**
I understand. You want to register and classify new contacts and automate the sales follow-up. May I have your name, please?

**USER**
Laura Gonzalez.

**ASSISTANT**
Thank you, Laura. What is the name of your business or professional activity?

**USER**
Impulso Digital.

**ASSISTANT**
Perfect, Laura. Do you have an approximate timeframe in mind for launching this automation?

**USER**
In two weeks.

# Conversation translation - live availability

The assistant validates the email, understands a natural date preference and offers only Calendar-backed options.

**ASSISTANT**
Understood. You would like to move forward in about two weeks. Could you provide your email address so I can send the meeting confirmation?

**USER**
l***@example.com

**ASSISTANT**
Can you confirm that your email address is l***@example.com?

**USER**
Yes.

**ASSISTANT**
Thank you, Laura. Which day would work best for the discovery meeting? You can give me a specific date or a time window, such as Monday morning or any afternoon.

**USER**
Monday, first thing in the morning.

**ASSISTANT**
Let me check the calendar.

**TOOL**
Availability result: status available.

**ASSISTANT**
On Monday, September 14, 2026, I have these morning options available: 10:30, 12:00 or 13:30. Which do you prefer for the 30-minute meeting?

**USER**
Could it be at 10:00?

**ASSISTANT**
10:00 is not available, but 10:30, 12:00 and 13:30 are. Would any of those work for you?

**USER**
10:30.

# Conversation translation - booking confirmation

The assistant summarizes the complete request, waits for a clear yes and only then calls the booking workflow.

**ASSISTANT**

Perfect, Laura. I will summarize the appointment before booking:

Name: Laura Gonzalez
Email: l***@example.com
Business: Impulso Digital
Service: lead capture and follow-up automation
Objective: register and classify contacts and automate sales follow-up
Date: Monday, September 14, 2026
Time: 10:30
Duration: 30 minutes

Would you like me to confirm the booking for that date and time?

**USER**

Yes.

**ASSISTANT**

Perfect. I will check the time again and confirm the booking.

**TOOL**

Booking result: status booked.

**ASSISTANT**

The meeting is booked for Monday, September 14, 2026 at 10:30, for 30 minutes. You will receive the email invitation, and the Google Meet link is included. Is there anything else you need?

**USER**

No.

**ASSISTANT**

Thank you for your call, Laura. The meeting about lead capture and follow-up automation is confirmed for Monday, September 14 at 10:30. Have a great day.

### What the conversation proves

Natural-language qualification, email verification, live availability, rejection of an unavailable request, explicit approval and a successful booked response all work in one continuous journey.